MPG No. 5.4.15 $2/m'$ [Type III, monoclinic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & T_y & 0 \end{bmatrix}$$

$$T_y = \mathbb{T}_y^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & T_{yyy} & 0 \\ 0 & T_{zzy} & 0 \\ T_{yzx} & 0 & T_{zzy} \\ 0 & T_{yzx} & 0 \\ T_{xxy} & 0 & T_{yzx} \end{bmatrix}$$

$$\begin{aligned} T_{xxy} &= \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} - \mathbb{T}_y^{\beta(1)} \\ T_{yyy} &= 3\mathbb{T}_y^{(1)} + 2\mathbb{T}_y^{\alpha(1)} \\ T_{zzy} &= \mathbb{T}_y^{(1)} - \mathbb{T}_y^{\alpha(1)} + \mathbb{T}_y^{\beta(1)} \\ T_{yzx} &= \mathbb{T}_{xyz}^{(1)} \end{aligned}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & T_{zzy} & 0 \\ T_{yzx} & 0 & -\frac{T_{zzy}}{2} \\ 0 & T_{zxy} & 0 \\ -\frac{T_{xxy}}{2} & 0 & -T_{yzx} - T_{zxy} \end{bmatrix}$$

$$\begin{aligned} T_{xxy} &= -2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)} \\ T_{zzy} &= 2\mathbb{M}_{xz}^{(1)} + 2\mathbb{T}_y^{(2)} \\ T_{yzx} &= 3\mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)} \\ \frac{T_{zzy}}{2} &= \mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)} \\ T_{zxy} &= -3\mathbb{M}_u^{(1)} + \mathbb{M}_v^{(1)} \\ \frac{T_{xxy}}{2} &= -\mathbb{M}_{xz}^{(1)} + \mathbb{T}_y^{(2)} \end{aligned}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} T_{yzx} & 0 & T_{yzz} \\ 0 & T_{zxy} & 0 \\ T_{xyx} & 0 & T_{xyz} \end{bmatrix}$$

$$\begin{aligned} T_{yzx} &= \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(2)} + \mathbb{M}_v^{(2)} \\ T_{yzz} &= \mathbb{M}_{xz}^{(2)} - \mathbb{T}_y^{(3)} \\ T_{zxy} &= \mathbb{M}_0^{(1)} - \mathbb{M}_u^{(2)} - \mathbb{M}_v^{(2)} \\ T_{xyx} &= \mathbb{M}_{xz}^{(2)} + \mathbb{T}_y^{(3)} \\ T_{xyz} &= \mathbb{M}_0^{(1)} + 2\mathbb{M}_u^{(2)} \end{aligned}$$

* Rank 4 tensor (sss). * Rank 4 tensor (ssa). * Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa). * Rank 4 tensor (as). * Rank 4 tensor (s). * Rank 4 tensor (a). * Rank 4 tensor (t).